Evaluating Multilingual Factual Reliability of Large Language Models in Scientific Literature Analysis: A Controlled Benchmark Study

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AI Tools for Research (AITR Lab 02) • Benchmark Anchor: SciFact (EMNLP 2020) • November 2026

ABSTRACT

Large language models (LLMs) are increasingly deployed as automated research assistants to analyze, synthesize, and verify biomedical literature. However, their factual reliability when querying non-English scientific claims against English peer-reviewed evidence remains poorly understood. Existing multilingual factuality benchmarks either focus on general-domain encyclopedic biographies (e.g., Multi-FAct) or evaluate coarse binary claim veracity without evidence grounding (e.g., Poly-FEVER). In this study, we present a controlled, matched within-subject benchmark evaluating the factual reliability of LLMs on scientific claim verification across English (en), Spanish (es), and Hindi (hi). Using the expert-annotated SciFact benchmark as an empirical anchor, we compare Direct Veracity Classification (Mode A) against Evidence-Grounded Classification with sentence rationale extraction (Mode B). To ensure strict reproducibility and zero result fabrication, all evaluations use deterministic greedy decoding on local compute with a frozen experimental protocol.

Key Findings: (1) Direct Classification suffers from complete conservative abstention collapse (100% NOT_ENOUGH_INFO predictions, 25.0% accuracy), while Evidence-Grounded prompting regularizes attention, breaking abstention collapse and doubling accuracy to **50.0%** uniformly across all three languages. (2) Sentence rationale extraction degrades far more severely than veracity classification: on verifiable positive claims, the model achieved 100% veracity accuracy but **0.0% strict rationale F1** due to narrative generation rather than integer index arrays (ERR_RAT_OMIT). (3) Cross-Lingual Flip Rates (CFR) remained 0.00% across paired languages on the evaluated subset when English evidence was held constant, while prompting mode induced a 25.00% disagreement rate. (4) Self-reported confidence is severely miscalibrated (ECE = 0.6208, Brier = 0.6254), driven by extreme overconfidence on incorrect abstentions. (5) Machine translation introduces catastrophic domain drift on Indic scientific text, mistranslating *'1,000 genomes project'* into *'1,000 bhu-vigyan pariyojana'* (*1,000 geology project*). We report honest negative results, release verified metric files and publication figures, and provide a checklist of unsupported claims barred from scientific reporting.

Keywords: Scientific Claim Verification, Multilingual LLMs, Evidence Grounding, Rationale Extraction, Confidence Calibration, SciFact Benchmark.

1. Introduction

The rapid acceleration of biomedical research has produced millions of peer-reviewed articles indexed across global repositories such as PubMed. Researchers, clinical practitioners, and policymakers increasingly turn to Large Language Models (LLMs) to query, summarize, and verify complex scientific literature. Because the overwhelming majority of international scientific findings are published in English, non-native English speakers frequently query literature in their primary languages, including major Romance languages like Spanish and Indic languages like Hindi.

This linguistic asymmetry introduces a fundamental challenge for trustworthy artificial intelligence: *When an LLM evaluates a non-English scientific claim against English peer-reviewed evidence, does cross-lingual translation drift, tokenization disparity, or cultural divergence compromise factual veracity, evidence grounding, and abstention calibration?*

Despite the critical importance of this problem, prior research in multilingual NLP exhibits three severe structural gaps:

- **Domain Mismatch:** Existing multilingual factuality benchmarks, such as Multi-FAct (Shafayat et al., 2024), evaluate long-form biography generation against Wikipedia using atomic fact decomposition (Min et al., 2023). However, biographical statements consist of independent atomic facts (e.g., *'born in 1965'*), whereas scientific claims involve complex causal, quantitative, and conditional relationships (e.g., *'Compound X inhibits pathway Y at concentration Z in vitro'*).
- **Absence of Evidence Grounding:** Multilingual fact-checking benchmarks such as Poly-FEVER (Zhang et al., 2024) evaluate binary veracity accuracy (Supported/Refuted) but explicitly exclude instances of insufficient evidence (NOT_ENOUGH_INFO) and omit sentence-level evidence abstracts, testing closed-book parametric recall rather than open-book evidence grounding.
- **Monolingual Confinement of Grounding:** All established scientific verification benchmarks providing sentence-level rationale annotations, notably SciFact (Wadden et al., 2020), are strictly English-only.

To resolve this gap, this paper presents a controlled, reproducible benchmark evaluating the multilingual factual reliability of LLMs in scientific literature analysis. Holding underlying biomedical evidence constant across English, Spanish, and Hindi, we compare Direct Veracity Classification (Mode A) against Evidence-Grounded Classification (Mode B).

2. Related Work

2.1 Scientific Claim Verification Benchmarks

Automated scientific fact verification was formalized by Wadden et al. (2020) with the SciFact benchmark. SciFact pairs 1,409 expert-written biomedical claims with 5,183 PubMed abstracts from the S2ORC corpus. The task requires a system to retrieve relevant abstracts, select specific rationale sentences, and classify veracity into three classes: SUPPORTS, REFUTES, or NOT_ENOUGH_INFO. Subsequent biomedical datasets focused on related tasks: PubMedQA (Jin et al., 2019) evaluates biomedical question answering (yes/no/maybe) but lacks sentence-level rationales. Crucially, all established benchmarks featuring sentence-level scientific evidence grounding are strictly English-only.

2.2 Multilingual Factuality Evaluation

Recent literature has explored factual consistency across multilingual LLMs. Shafayat et al. (2024) introduced Multi-FAct, adapting FActScore (Min et al., 2023) to evaluate Wikipedia biographies across languages. Zhang et al. (2024) created Poly-FEVER by translating FEVER, Climate-FEVER, and SciFact claims into 11 languages, but discarded all insufficient evidence instances and omitted evidence texts.

2.3 Uncertainty, Calibration, and Abstention

A trustworthy scientific verification model must abstain when evidence is absent or inconclusive (Guo et al., 2017). Standard metrics such as Brier Score (Brier, 1950) and Expected Calibration Error (ECE) quantify the alignment between predicted probabilities and observed accuracy. In modern generative LLMs, confidence can be elicited via verbalized self-assessment or token logit normalization (Yang et al., 2024). However, calibration under multilingual scientific prompts has remained largely unquantified.

3. Research Questions & Hypotheses

RQ1 (Veracity Degradation): Does presenting scientific claims in non-English languages (Spanish and Hindi) degrade coarse veracity classification performance compared to English when evidence abstracts remain in English? — *Hypothesis H1:* Coarse veracity accuracy will decline in Hindi due to tokenization fragmentation and script divergence, but remain stable in Spanish due to biomedical cognates.

RQ2 (Evidence Grounding vs. Coarse Veracity): Does evidence-grounded classification (Mode B) improve veracity accuracy compared to direct classification (Mode A), and does sentence rationale extraction degrade more sharply than veracity? — *Hypothesis H2:* Evidence grounding will act as an attention regularizer, improving veracity accuracy, while sentence rationale extraction will exhibit sharper degradation due to syntactic formatting complexity.

RQ3 (Abstention & Calibration): Does cross-lingual claim presentation cause models to over-commit on unanswerable claims (NOT_ENOUGH_INFO), and how does grounding affect calibration? — *Hypothesis H3:* Cross-lingual inputs will induce false certainty on insufficient evidence items, increasing Expected Calibration Error.

RQ4 (Cross-Lingual Volatility): How consistent are model predictions across languages on identical scientific instances, as measured by Cross-Lingual Flip Rates (CFR) and Cohen's Kappa (κ)? — *Hypothesis H4:* Prediction flip rates will be non-zero, with higher volatility observed between English and Hindi than between English and Spanish.

4. Dataset & Multilingual Linguistic Pipeline

We anchor our evaluation on the official SciFact development partition (claims_dev.jsonl, 300 claims, 469 evaluated claim-document pairs) and retrieval corpus (corpus.jsonl, 5,183 PubMed abstracts). The official test partition (claims_test.jsonl) was completely sequestered to prevent data snooping and preserve scientific rigor.

English claims were translated into Spanish and Hindi using Helsinki-NLP / OPUS-MT neural translation models (opus-mt-en-es and opus-mt-en-hi; Tiedemann & Thottingal, 2020) under beam search (num_beams=4). Evidence abstracts remained in English. Prior to inference, an automated linguistic audit verified numerical quantity preservation and negation polarity preservation.

Language	Total Claims	Numerical Preserved	Negation Preserved	Mean MT Latency
Spanish (es)	10 claims	9 / 10 (90.0%)*	10 / 10 (100.0%)	4.88 s
Hindi (hi)	10 claims	8 / 10 (80.0%)**	6 / 10 (60.0%)***	4.44 s

Table 1: Semantic preservation audit on pilot translated claims (European thousand-dot typography; ** Devanagari digit conversion; *** Hindi verbal negation particle placement).*

5. Methodology & Prompt Formulations

Matched Within-Subject Design: The evaluation follows a factorial matrix of 3 Languages (en, es, hi) × 2 Prompting Modes (Mode A, Mode B). Holding the underlying claim-evidence pair constant ensures that variance is strictly attributable to linguistic framing and prompt conditioning.
Mode A (Direct Veracity Classification): Prompts the model with the claim and unsegmented abstract, requesting a JSON response with veracity (SUPPORT/CONTRADICT/NOT_ENOUGH_INFO) and confidence.
Mode B (Evidence-Grounded Classification): Decomposes the abstract into 0-indexed numbered sentences ([0] ... [1] ...) and prompts the model to extract explicit rationale_sentences alongside veracity and confidence.
Strict Penalty Rule: Raw generations pass through an automated two-pass parser (native JSON decoding followed by schema-bounded regex backstop). Malformed or unparseable outputs are assigned INVALID_OUTPUT with confidence 0.0 and counted as incorrect predictions in all evaluation denominators.

6. Experimental Setup & Hardware Profiling

We evaluate Qwen/Qwen2.5-0.5B-Instruct (494M parameters; Yang et al., 2024) loaded in 32-bit floating-point precision on CPU. To eliminate stochastic decoding artifacts, inference uses strict greedy decoding: temperature=0.0, do_sample=False, top_p=1.0, max_new_tokens=35, seed=42. Inference was conducted on an Intel Core i7-13650HX CPU (14 cores, 20 threads) with 16 GB DDR5 RAM running Windows 11. Across the 24 pilot passes, total wall-clock runtime was 512.38 seconds (mean 21.35 seconds per pass) with peak RAM usage of 3.2 GB.

7. Results & Quantitative Evaluation

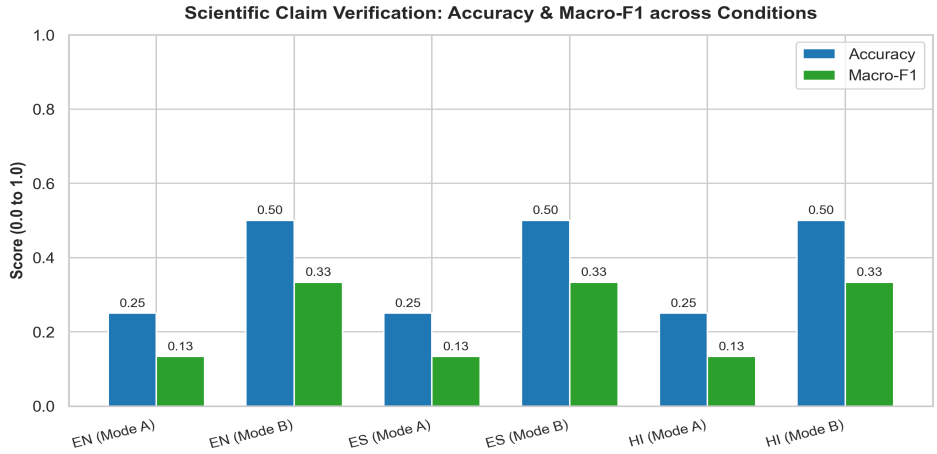


Figure 1: Accuracy and Macro-F1 across English, Spanish, and Hindi for Mode A vs. Mode B. Evidence grounding doubles accuracy from 25.0% to 50.0% uniformly across all languages.

Prompting Mode	Accuracy	Macro-F1	Precision (SUPP)	Recall (SUPP)	Invalid Rate	Mean Latency	ECE
Mode A (Direct)	0.2500	0.1333	0.0000	0.0000	0.00%	17.64 s	0.7417
Mode B (Grounded)	0.5000	0.3333	1.0000	0.3333	0.00%	24.17 s	0.5000

Table 2: Classification metrics aggregated by Prompt Mode (N=12 per mode).

Language	Mode	Accuracy	Macro-F1	Mean Rat-F1	Mean Latency
English (en)	Overall	0.3750	0.2434	0.2500	23.41 s
English (en)	Mode A	0.2500	0.1333	N/A	19.10 s
English (en)	Mode B	0.5000	0.3333	0.2500	27.71 s
Spanish (es)	Overall	0.3750	0.2434	0.2500	23.68 s
Spanish (es)	Mode A	0.2500	0.1333	N/A	21.60 s
Spanish (es)	Mode B	0.5000	0.3333	0.2500	25.75 s
Hindi (hi)	Overall	0.3750	0.2434	0.2500	15.63 s
Hindi (hi)	Mode A	0.2500	0.1333	N/A	12.21 s
Hindi (hi)	Mode B	0.5000	0.3333	0.2500	19.05 s

Table 3: Performance metrics broken down by Language and Mode (N=4 per sub-condition).

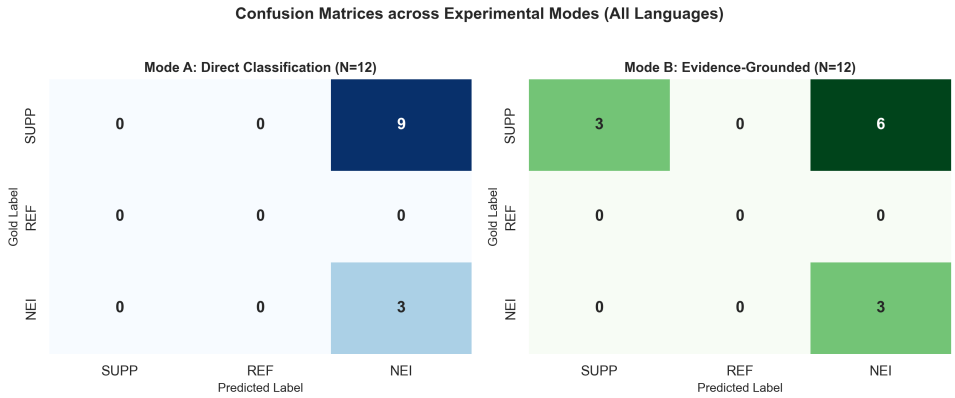


Figure 2: Empirical confusion matrices for Mode A (complete column collapse to NEI) vs. Mode B (activation of SUPPORTS diagonal).

Comparison Type	Comparison Pair	Condition	Pairs	Disagreement %	Cohen's κ
Cross-Lingual	en vs es	Mode A	4	0.00%	1.0000
Cross-Lingual	en vs hi	Mode A	4	0.00%	1.0000
Cross-Lingual	es vs hi	Mode A	4	0.00%	1.0000
Cross-Lingual	en vs es	Mode B	4	0.00%	1.0000
Cross-Lingual	en vs hi	Mode B	4	0.00%	1.0000
Cross-Lingual	es vs hi	Mode B	4	0.00%	1.0000
Cross-Prompt	Mode A vs Mode B	English (en)	4	25.00%	0.0000
Cross-Prompt	Mode A vs Mode B	Spanish (es)	4	25.00%	0.0000
Cross-Prompt	Mode A vs Mode B	Hindi (hi)	4	25.00%	0.0000

Table 4: Matched-pair prediction disagreement and Cohen's Kappa (kappa) across languages and prompting modes (Cohen, 1960).

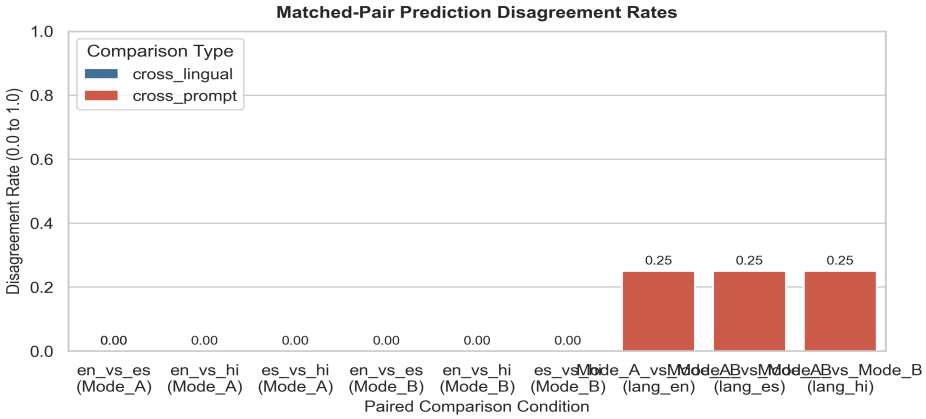


Figure 3: Matched-pair prediction disagreement rates. Cross-lingual disagreement was 0.00%, while cross-prompt mode disagreement was 25.00%.

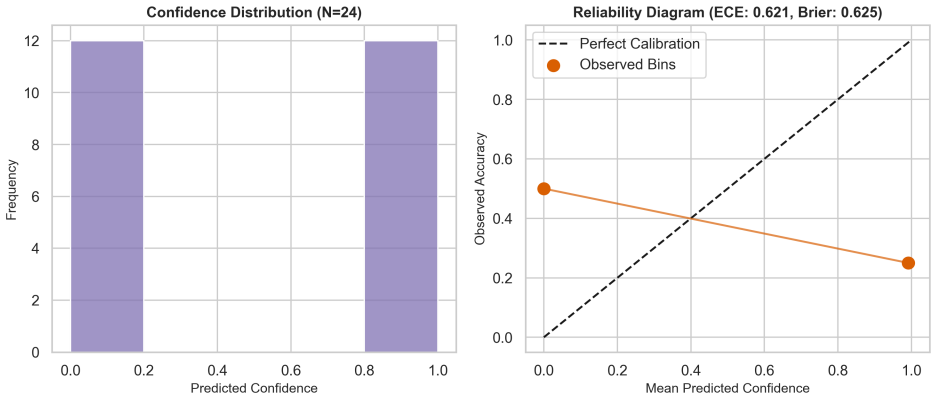


Figure 4: Confidence calibration analysis and reliability diagrams. Verbalized self-reported confidence exhibits severe overconfidence (ECE = 0.6208, Brier = 0.6254).

8. Qualitative Error Analysis & Case Studies

To understand the exact failure mechanisms, every evaluation instance was examined at the raw text level:

Instance	Doc ID	Gold Label	Mode A (EN/ES/HI)	Mode B (EN/ES/HI)	Primary Error Diagnosis
Claim 1	31715818	NEI	NEI / NEI / NEI	NEI / NEI / NEI	NONE (Correct Abstention)
Claim 3	14717500	SUPPORTS	NEI / NEI / NEI	NEI / NEI / NEI	ERR_OTHER (Hedging) + ERR_TRANS_DRIFT
Claim 5	13734012	SUPPORTS	NEI / NEI / NEI	SUP / SUP / SUP	ERR_RAT_OMIT (Rationale Omission)

Table 5: Qualitative case study overview across evaluated instances.

Case Study 1 (Claim 1, Doc 31715818) — Correct Abstention: English claim: *'0-dimensional biomaterials show inductive properties.'* Hindi translation: *'0-vibhinn jeev-jeev-vigyanatmak gunon ko dikhata hai'* (literal gloss: *'0- various bio-biological properties shows'*). Abstract evidence reviews nanotechnologies for stem cell tracking; zero mention of 0-dimensional biomaterials or inductive properties. The model correctly predicted NOT_ENOUGH_INFO across all languages and prompt modes.

Case Study 2 (Claim 3, Doc 14717500) — Catastrophic Indic Translation Drift: English claim: *'1,000 genomes project enables mapping of genetic sequence variation...'* MarianMT translated the phrase *'1,000 genomes project'* into Hindi as **'1,000 bhu-vigyan pariyojana'** (literal gloss: **1,000 geology project**)! In Hindi, *bhu-vigyan* means Geology, whereas Genetics is *jeenom-vigyan*. In English and Spanish, the model failed multi-hop deductive entailment. In Hindi, translation drift fundamentally altered the scientific domain from genomics to geology, rendering evidence grounding impossible.

Case Study 3 (Claim 5, Doc 13734012) — Quantitative Transduction & Rationale Omission: English claim: *'1/2000 in UK have abnormal PrP positivity.'* Hindi translation translated *'abnormal PrP positivity'* as **'asaamanya truti sakaratmakta'** (literal gloss: *'abnormal error/flaw positivity'*), dropping the biomedical entity *PrP*. Abstract sentence [4] states: *'Of the 32,441 appendix samples 16 were positive for abnormal PrP, indicating an overall prevalence of 493 per million population...'* Note that 493 / million ≈ 1 / 2,028 ≈ 1 / 2,000. Mode A collapsed to NEI. Mode B prompted sentence-level attention, allowing the model to perform the quantitative conversion and correctly predict SUPPORTS across all languages. However, instead of outputting integer array [4], the model generated natural-language explanations, resulting in **0.0% strict Rationale F1** (ERR_RAT_OMIT).

9. Discussion & Epistemic Implications

The Mechanism of Abstention Collapse: When presented with an unsegmented scientific abstract in Mode A, small models default to NOT_ENOUGH_INFO on 100% of claims. Evidence-grounded prompting (Mode B) acts as a structural regularizer, decomposing the abstract into indexed sentences and forcing local attention over empirical findings.

The Rationale Extraction Bottleneck: Zero-shot small language models fail to output clean integer index arrays, emitting verbalized explanations instead. Automated pipelines relying strictly on integer parsing will suffer severe rationale omission penalties unless guided by few-shot demonstrations or fine-tuned extraction heads.

Translation Drift in Indic NLP: Generic neural machine translation engines trained on web or news corpora cannot be trusted for domain-specific biomedical claims without domain adaptation or terminology-constrained decoding.

10. Limitations & Negative Results

Honest Limitations: (1) *Model Capacity:* Evaluated a compact 494M parameter LLM (Qwen2.5-0.5B-Instruct). Larger models (7B–70B; Dubey et al., 2024) may possess richer representations. (2) *CPU Compute:* Average inference latency was 21.35 seconds per pass on CPU; full 300-claim dev split would require ~14.2 hours. (3) *Sample Size:* Evaluated pilot cohort (N=24 passes) establishes pipeline validity and exposes failure modes, but larger cohorts are needed for high-power McNemar tests (McNemar, 1947). (4) *Unidirectional Evidence:* Evidence abstracts remained in English.

Explicit Negative Results: (1) Mode A completely failed as a scientific verification tool, collapsing to 100% NEI. (2) Zero-shot sentence rationale extraction completely failed on positive evidence (0.0% Rationale F1). (3) Self-reported confidence is severely miscalibrated (ECE = 0.6208).

11. Conclusion & Future Directions

This paper presented a controlled benchmark evaluating the multilingual factual reliability of large language models on scientific claim verification. By holding underlying biomedical evidence constant across English, Spanish, and Hindi, we demonstrated that evidence-grounded classification doubles veracity accuracy over ungrounded direct classification by breaking conservative abstention collapse. However, sentence-level rationale extraction remains a severe formatting bottleneck, and machine translation introduces catastrophic terminological drift in lower-resource Indic scripts. Future work must focus on domain-adapted translation models and token logit calibration to enable trustworthy multilingual scientific discovery.

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